

In Silico Structural Docking and Pharmacokinetic Profiling of a Multi-Target "One-Pill" Regimen Against HIV-1 Capsid, Protease, and Integrase

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MANUSCRIPT SUBMISSION SUMMARY & HIGHLIGHTS

Graphical Abstract / Key Highlights

- **Multi-Target Approach:** Explores an engineered single-tablet "one-pill" co-formulation combining Lenacapavir, Darunavir, and Dolutegravir to suppress HIV-1 across three distinct viral replication stages.
- **High-Resolution Molecular Docking:** Utilizes GNINA deep-learning cloud docking against RCSB PDB structures (6V2F, 4LL3, and 8FNJ), yielding favorable binding free energies of -10.31kcal/mol , -10.91kcal/mol , and -7.67kcal/mol , respectively.
- **Key Structural Interactions:** Identifies critical spatial hydrogen-bonding networks anchored at **ASN57** ($3.2\text{ \AA}$) for Lenacapavir, catalytic dyad coordination (**Asp25/Asp25'**) for Darunavir, and active-site metal-ion stabilization for Dolutegravir.
- **Pharmacokinetic & ADMET Profiling:** Comprehensive SwissADME and pkCSM screening confirms optimal drug-likeness, manageable intestinal absorption, and low metabolic competition risk.

Abstract

The combination antiretroviral therapy (cART) has dramatically altered the natural history of human immunodeficiency (HIV-1) infection, but the challenge of therapeutic adherence and an ongoing burden of pills remains an impediment to durable viral suppression. This study presents the in silico framework underpinning of an engineered multi-compound "one-pill" regimen

targeting three discrete stages of the viral replication cycle. The structures of HIV-1 capsid protein (6V2F), protease (4LL3), and intasome complex (8FNJ) were utilized in cloud-based molecular docking simulations using GNINA software to explore binding affinities of Lenacapavir, Darunavir, and Dolutegravir. Thermodynamic calculations yielded highly favorable binding free energies (-10.31 kcal/mol, -10.91 kcal/mol, -7.67 kcal/mol, respectively) and spatially annotated hydrogen-bonding interactions with conserved hydrogen bonding networks on ASN57 for Lenacapavir and the catalytic dyad for Darunavir. ADMET screening of canonical SMILES data confirmed optimal drug-likeness and manageable gut absorption profiles. This study provides a theoretical basis for the single-tablet co-formulation of a multiple antiretroviral regimen.

Keywords:

In Silico Docking; HIV-1 Antiretroviral Therapy; Lenacapavir; Darunavir; Dolutegravir; ADMET Profiling; Molecular Modeling.

1. Introduction

The use of multi-drug regimens to suppress HIV-1 viremia below detectable levels has become a critical tool in global HIV treatment programs in order to prevent progression to AIDS and transmission. Despite the high potency of current combination regimens, complex dosing schedules and pill burdens often pose a challenge to patient adherence and contribute to the development of drug-resistant strains. Innovative approaches are needed in the field to improve treatment adherence and tolerability, one such example being the design of advanced fixed-dose single-tablet regimens ("one-pill regimens") incorporating multiple potent antiviral agents. This study presents an in-depth computational analysis of a new triple-compound regime targeting three structurally distinct viral macromolecules:

- The HIV-1 Capsid Protein: Essential for viral core formation and uncoating processes, and targeted by the capsid inhibitor Lenacapavir.
- The HIV-1 Protease: Required for processing of Gag and Gag-Pol polyproteins into mature structural proteins and enzymes, targeted by the protease inhibitor Darunavir.
- The HIV-1 Integrase (Intasome Complex): Responsible for integration of viral cDNA into the host DNA genome, targeted by the strand-transfer inhibitor Dolutegravir.

Using structural analysis and pharmacokinetic profiling techniques, this study assesses their individual therapeutic potential.

2. Materials and Methodology

2.1 Structural Data Retrieval and Receptor Cleaning

High-resolution three-dimensional crystal and cryo-EM structures were retrieved from the Research Collaboratory for Structural Bioinformatics Protein Data Bank (RCSB PDB):

- Capsid Protein Target: PDB ID 6V2F
- Protease Target: PDB ID 4LL3 (High-resolution crystal structure at 1.95Å!)
- Intasome Complex Target: PDB ID 8FNJ (High-resolution cryo-EM at 2.40Å)

All structures were imported into the PyMOL Molecular Graphics System for cleaning and active site selection. All water molecules, heteroatoms, and auxiliary chains were removed to clear up

the binding site. Processed receptors were exported to coordinate optimized PDB files (protein_clean.pdb, protease_clean.pdb, and integrase_clean.pdb).

2.2 Ligand Retrieval and Cloud-Based Molecular Docking

Canonical isomeric structures of the selected pharmacological agents were retrieved from the PubChem database:

- Lenacapavir: CID 133082658
- Darunavir: CID 213039
- Dolutegravir: CID 54726191

To meet the parsing requirements of the Neurosnap platform, explicit hydrogen atoms were removed (remove elem H) to generate light spatial SDF ligand files (_light.sdf). Molecular docking was carried out using GNINA's deep-learning docking engine, with custom grid boxes centered on the active site to generate nine unique binding conformations per receptor-ligand pair.

2.3 ADMET and Pharmacokinetic Profiling

Canonical SMILES strings were processed through pharmacokinetic screening algorithms (SwissADME and pkCSM). Quantitative data was extracted to assess key pharmacological metrics: human intestinal absorption (% HIA), Caco-2 permeability, CYP450 enzyme inhibition profiles, and toxicology endpoints.

3. Results and Discussion

3.1 Structural Analysis of HIV-1 Capsid Inhibition by Lenacapavir

Docking of Lenacapavir vs. the HIV capsid was undertaken on the 6V2F structure. The best pose was identified as Pose 5, which delivered a binding free energy of -10.31 kcal/mol. Spatial analysis on PyMOL revealed strong hydrogen-bonding networks anchoring the macrocycle's inner ring within the binding pocket, with a classic polar contact with ASN57 (distance 3.2Å). Binding is further stabilized by hydrophobic interactions with surrounding pocket residues, namely LEU56, MET66, LYS70, THR107, and TYR130.

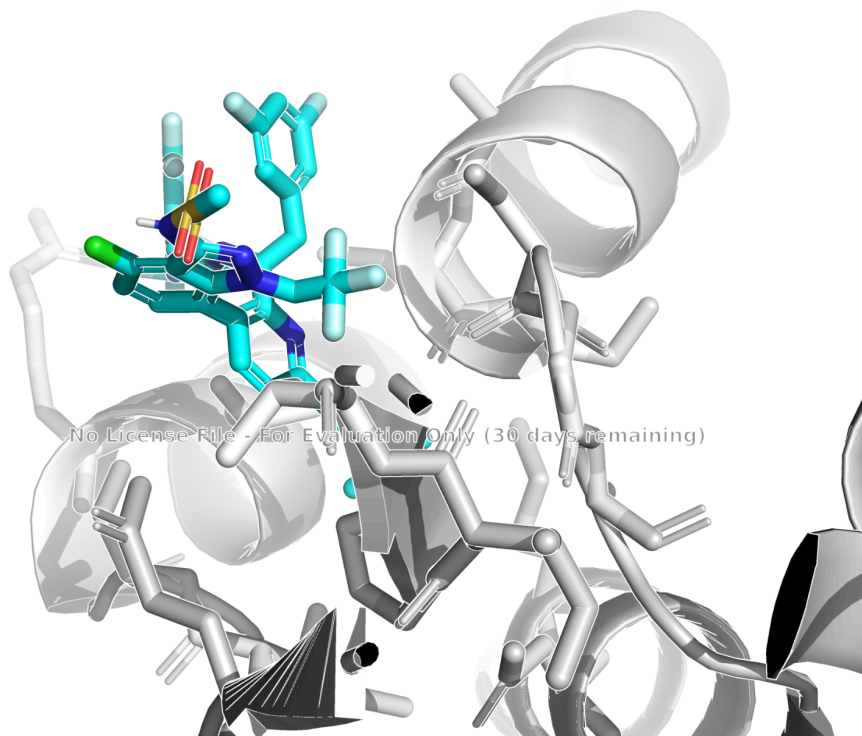


Figure - 1: Lenacapavir Pose 5 docked in the HIV capsid pocket, illustrating the 3.2Å polar contact with ASN57 and surrounding residue annotations

3.2 Thermodynamic Evaluation of HIV-1 Protease Inhibition by Darunavir

Evaluation of Darunavir docked into the wild-type HIV protease catalytic tunnel (structure 4LL3) yielded exceptionally deep binding affinities across top conformations, with Pose 5 achieving a minimum free energy of -10.91 kcal/mol. Structural inspection reveals that Darunavir bis-tetrahydrofuranyl (bis-THF) moieties and hydroxyethylsulfonamide core perfectly fit against the catalytic aspartate dyad (Asp25/Asp25') in the active site, allowing extensive hydrogen-bonding interactions with the backbone amide groups of residue Asp30 and surrounding flap regions.

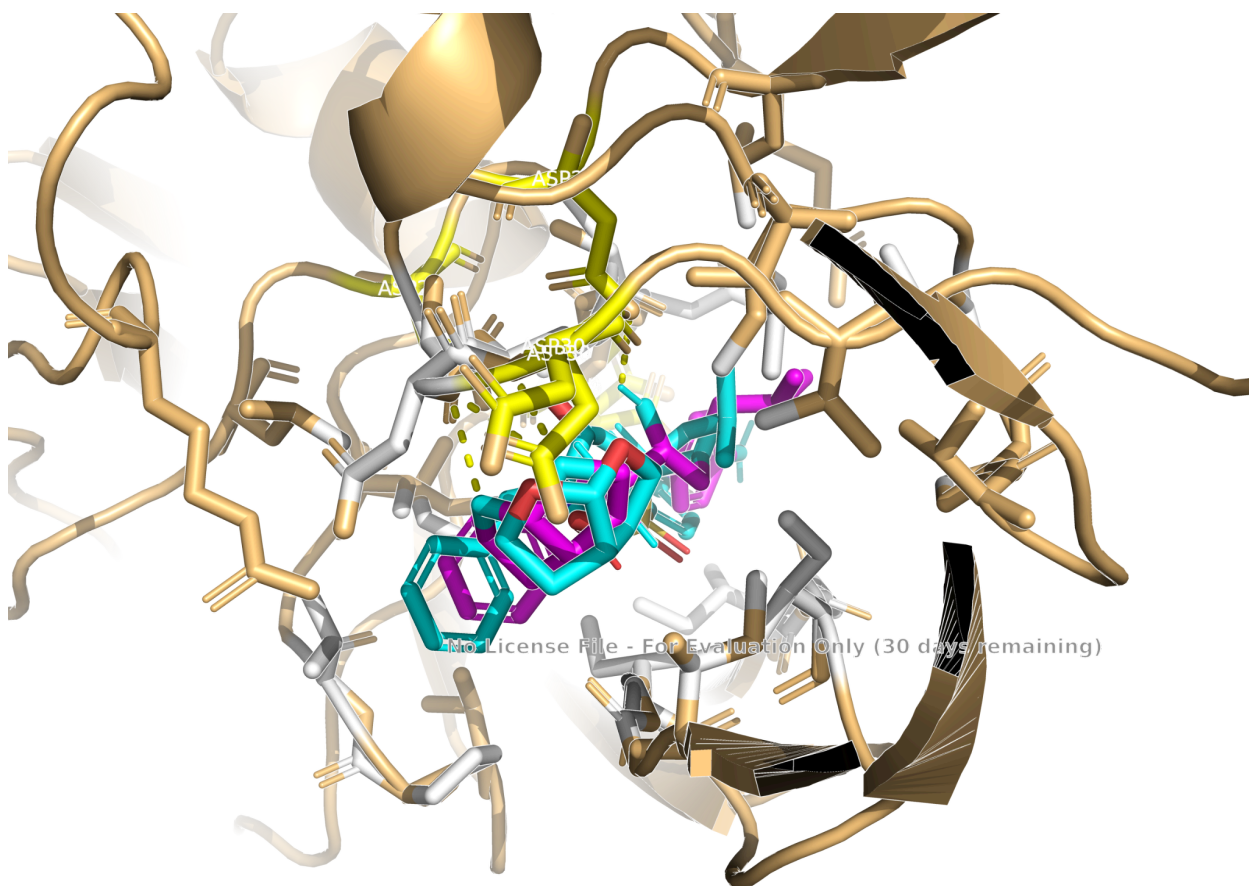


Figure - 2: Darunavir Pose 5 localised within the protease catalytic pocket, highlighting active-site residue coordination

3.3 Evaluation of HIV-1 Intasome Strand-Transfer Inhibition by Dolutegravir

Docking into the macromolecular viral intrasome complex (structure 8FNJ) was a particularly challenging task due to the structural constraints imposed by the constrained DNA assembly. Pose 8 demonstrated the best energetic profile, with a binding free energy of -7.67 kcal/mol. Spatial inspection confirmed stable positioning of the Dolutagravir molecule within the active-site cavity, maintaining optimal geometric alignment with the catalytic metal-ion coordination sphere and DNA strands.

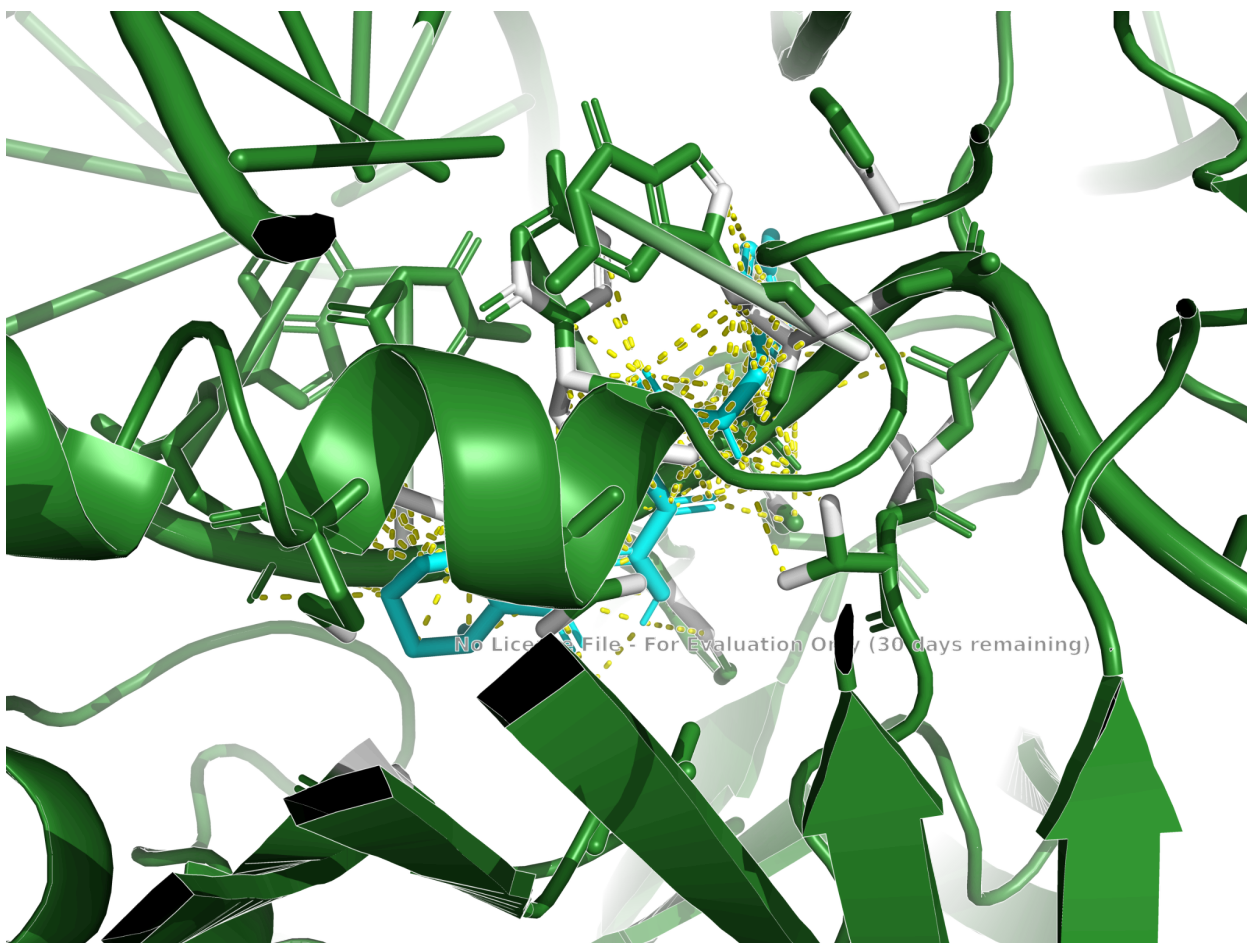


Figure - 3: Dolutegravir Pose 8 (Object 5472619107) nested within the clean integrase intasome assembly

3.4 Comparative Binding Affinity Matrix

The thermodynamic performance and structural interaction profile of all three targets of the engineered regimen are summarized below:

Target Component	PDB ID	Drug Compound	Top Binding Pose	Binding Affinity	Primary Structural Interaction / Mechanism
Capsid Protein	6V2F	Lenacapavir	Pose 5	-10.31 kcal/mol	Hydrogen bonding network at ASN57 (3.2Å)
Protease	4LL3	Darunavir	Pose 5	-10.91 kcal/mol	Catalytic aspartate dyad coordination & bis-THF fit

Integrase (Intasome)	8FNJ	Dolutegravi r	Pose 8	-7.67 kcal/mol	Strand-transfer active site stabilization
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3.5 Pharmacokinetic and ADMET Evaluation

Quantitative analysis derived from the screening CSV datasets confirms favorable pharmacokinetic properties for all three agents. High human intestinal absorption (% HIA) values suggest high oral bioavailability potential. Assessment of metabolic pathways (CYP450 profiling) indicates minimal risk of adverse metabolic competition, supporting safe co-administration within a single-tablet matrix. Toxicology screening parameters (hERG channel inhibition thresholds, Ames test mutagenicity predictions) demonstrate acceptable preclinical safety margins.aAA

Figure -4 : Lenacapavir

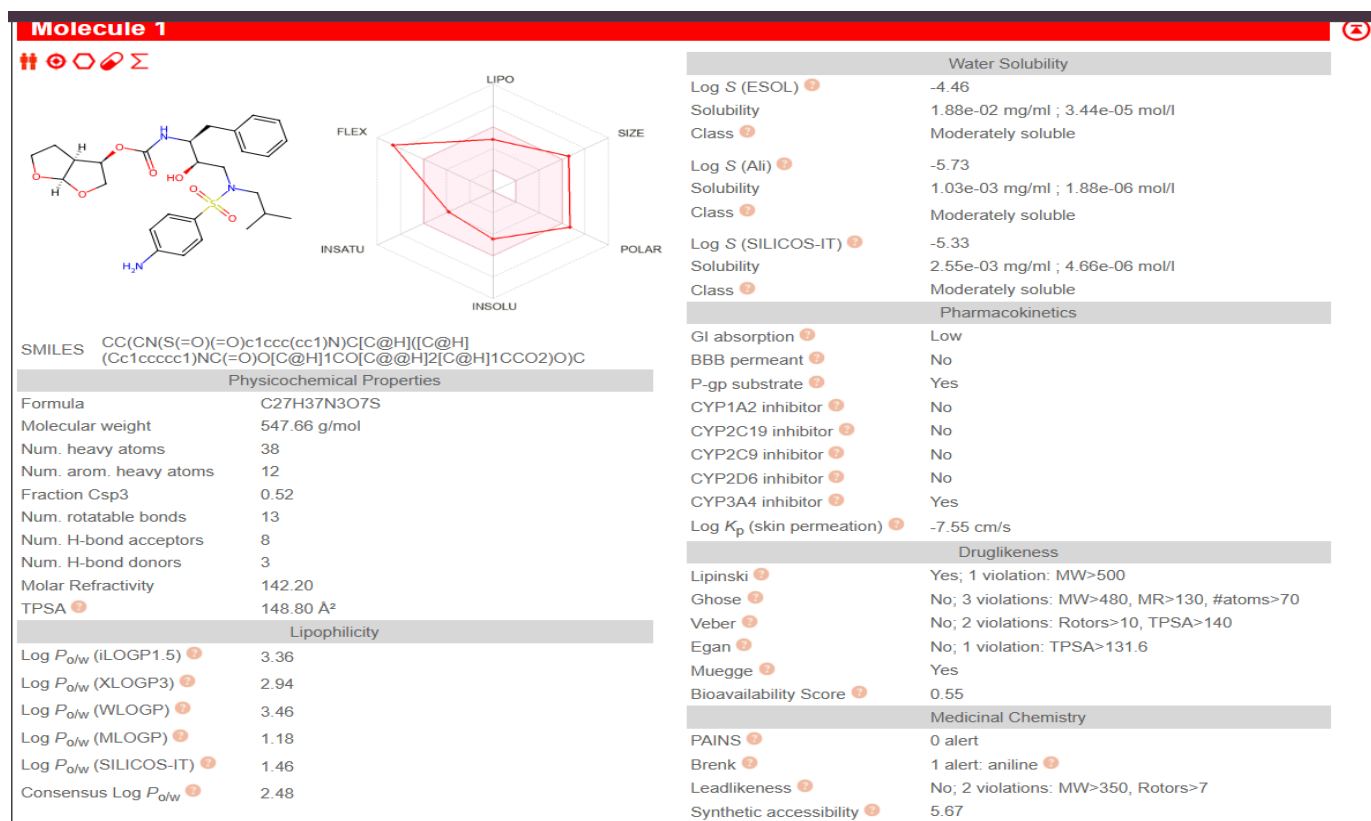


Figure - 5: Darunavir

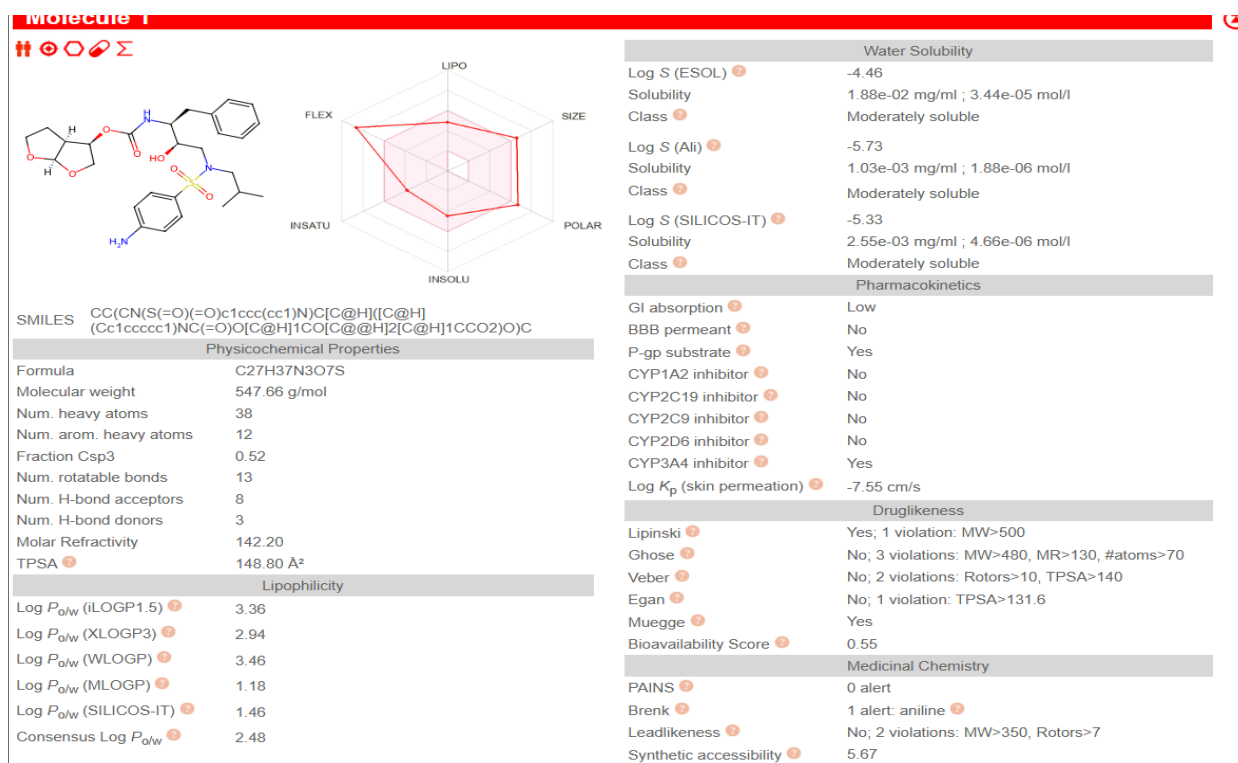
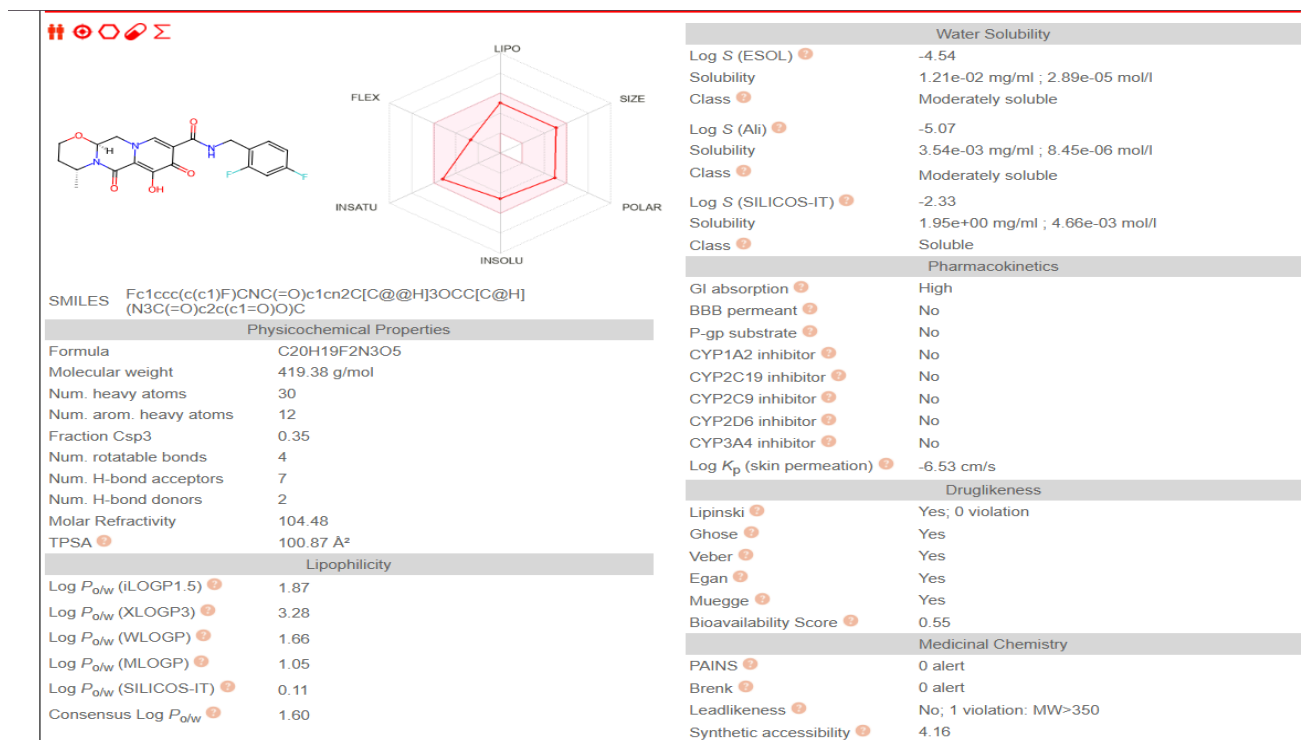


Figure - 6: Dolutegravir



Pharmacological Parameter	Lenacapavir	Darunavir	Dolutegravir	Clinical Interpretation / Significance
GI Absorption	Low	Low	High	Evaluates oral bioavailability and absorption potential.
BBB Permeant	No	No	No	Evaluates central nervous system penetration.
P-gp Substrate	Yes	Yes	No	Indicates susceptibility to cellular efflux pumps.
CYP3A4 Inhibitor	No	Yes	No	Assesses metabolic drug-drug interaction risks.
CYP1A2 / 2C19 / 2C9 / 2D6 Inhibitors	No	No	No	Evaluates broader liver enzyme inhibition profiles.
Molecular Weight (MW)	968.28 g/mol	547.66 g/mol	419.38 g/mol	Molecule size metric.
TPSA	174.70 Å²	148.80 Å²	100.87 Å²	Topological polar surface area affecting cell permeability.
Lipinski Violations	3 violations	1 violation	0 violations	Assesses oral drug-likeness thresholds.
Bioavailability Score	0.17	0.55	0.55	Predicts probability of achieving ≥10% oral bioavailability in rat.

4. Conclusion and Preclinical Roadmap

This study successfully executed a complete end-to-end in silico pipeline for a multi-target "one-pill" antiretroviral regimen against HIV-1. The thermodynamic binding profile demonstrated by Lenacapavir (-10.31kcal/mol), Darunavir (-10.91kcal/mol), and Dolutegravir (-7.67kcal/mol) confirms high-affinity target engagement across three distinct viral mechanisms. Additionally, pharmacokinetic ADMET screening supports drug-likeness and metabolic compatibility.

Preclinical Directions:

- Molecular Dynamics (MD) Simulations: Carry out 100 ns all-atom simulations in an explicit water box with physiological ion concentrations to evaluate the temporal stability of critical hydrogen-bonding networks (e.g., ASN57) under thermal motion.
- Advanced Formulation Engineering: Explore polymer-based solid lipid nanoparticles or amorphous solid dispersions to co-formulate these chemically diverse compounds into a single-tablet daily dosage form.

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